Rady | School of Management
UNIVERSITY OF CALIFORNIA SAN DIEGO

MGTF490: Capstone Applied Finance Project

Fall 2026, Master of Quantitative Finance

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OFFICE HOURS: to be announced, but instructors will be available to meet throughout the quarter. Just email us and make an appointment.

COURSE DESCRIPTION

The Master of Quantitative Finance Capstone course is a 4-unit course that involves an applied finance project where students will work in teams on a project that is associated with an industry partnership. Projects should enhance student knowledge of a particular area and increase student attractiveness to potential employers.

DELIVERABLE AND GRADING

A completed research paper is due no later than December 2. All 3 instructors will read each paper to determine a grade. Each team member must speak when giving team presentations.

Both presentation style and content will be evaluated. Content matters most but style is also important and good style will help the grade. Poor writing and style will detract. Professor Padernacht will provide guidance on speaking and writing, so teams may submit their presentation slides to her for advice prior to each presentation. All members of a team are expected to be present when giving their team's oral presentations.

FACTORS ENTERING INTO GRADE DETERMINATION

- Style: Correct spelling and grammar; attractive presentation style (very important in industry)
- Content:
  - Proper introduction and motivation of issue studied
  - Clear explanation of data, methods, and results
  - Originality of study (a novel analysis is a positive)
  - Technical skill: use and clear understanding of appropriate techniques

KEY DATES

- Class meets 9 am to noon or 1 to 4pm on the dates given below.
- First thing: All teams must prepare a 2-page project proposal (see **Guidelines** below)
- Presentation slides will be turned in for each class meeting on the EduSourced website (all students will be given access to this class management site)

- Sept 30: class meeting with instructors, research proposal due
  - Teams should turn in their project proposals and give a brief presentation of their project idea in class (no more than 7 minutes)

- Oct 21: class meeting with instructors

- Each team will give a brief progress update on the status of their project (no more than 7 minutes)

- Nov 18: class meeting with instructors
  - Each team will give a brief progress update on the status of their project (no more than 7 minutes)
  - In addition to your presentation slides, each team must turn in a 10-page outline of the research report, including any sections you have already written

- Dec 2: class meeting with instructors
  - Each team will give a 10 minute professional presentation on their research, stating the problem studied, the methods used, and the findings or results. The presentation must be concise and prepared with visual slides that are appropriate for a professional meeting.

- Dec 2 or before: Finished reports must be submitted to the EduSourced site. *The electronic copy must be in Word so that it can be edited as needed.*

HELPFUL HINTS AND WARNINGS

- Teams should consult frequently with the instructors
- Properly cite any relevant literature (plagiarism is a serious offense and ethical behavior is expected)
- Submit your computer code at the end so that the firm may replicate your study if they wish
- Others should be able to replicate your results—so be clear about your data and methods
  - If a proprietary data set is used, be sure to treat it appropriately—don't share with anyone outside your team
- Everyone on each team must make an adequate contribution (don't be a "free rider")
  - You will assess your teammates' contributions at the end of the class
- Meet all deadlines and show up for all class meetings

GUIDELINES FOR THE MASTER OF QUANTITATIVE FINANCE CAPSTONE PROJECT

The topics will be well-suited to being analyzed using quantitative techniques and for demonstrating your skills in using the methods taught in the Master of Quantitative Finance program. In the project, you must demonstrate your ability to formulate an interesting empirical problem, collect data to model the variables of interest, and analyze the data by means of appropriate methods. You are encouraged to use the empirical techniques covered in the program as a basis for your analysis and, if need be, explore more sophisticated methods.

Project Proposal
Before you start the project, you will have to prepare a two-page *project proposal*, addressing the following questions: (1) What is the main question of interest? (2) Why is it important? (3) What empirical and analytic methods would you use to answer the question? (4) What data would you need for your investigation? Do you have the data now? (5) What is the expected finding from the project? The project proposals will be approved by the instructors. If a proposal needs modifications, alterations, or additions, the faculty will communicate this to you. Once the proposal is approved, you will be ready to start working on the project itself. *The two-page project proposal should be turned in on EduSourced no later than Sept 30*.

Research Report

The page limit for the project itself is 50 pages including double-spaced text, figures, tables, references, appendices and supporting material. Make sure to use your own words throughout the project to describe everything, including ideas, models, results, and conclusions. Reports can be shorter, as is appropriate for the particular analysis. Each member of the team must write a section of the report and the author of each section should be listed.

OUTLINE OF PROJECT

Below is a sample project outline. The actual content of your project might vary and will depend on the topic at hand.

Project Title, Group Members

1. Introduction
   - Motivation: Why is this an interesting and challenging yet feasible empirical problem?
   - Brief review of existing work in the area (if relevant)
   - Summary of your main empirical findings
   - Outline of the remaining part of your project

2. Outline of the empirical question
   - Set up the empirical problem
   - How have others tackled the problem (if there's existing work on the topic)? This could serve as a benchmark model

3. Data
   - Data sources
   - Plot graphs of the key variables you will use
   - Data description: mean, st.dev, first-order autocorrelation (persistence) for all included variables
   - Explain any transformations you have used (e.g. log(x/x(-1)) – are you modeling growth rates or levels?
   - Comment on seasonal/trend/cyclical components if relevant
   - Did you use mixed data frequencies (daily/monthly/quarterly/annual data)? If so, how did you convert the series into a common frequency (e.g. monthly data)?
   - Did you merge data from different sources? Make sure to plot the merged data to ensure that they look similar before and after the merge.

4. Econometric Methods
   - Describe the types of econometric strategies/models you will consider (possibly only one approach)
   - Describe the estimation methods used (OLS, MLE, GMM, Kalman filtering, logit/probit etc).

5. Empirical Results
   - Results for the estimated model(s)
   - How did you arrive at the preferred model?
   - How precise is the model's in-sample fit (e.g. its R_2 or directional accuracy)?
   - Comment on t-statistics/p-values. Are the signs of the coefficients as expected?
   - Split-sample results: if regime breaks are relevant, report results demonstrating the interesting changes that occur.
   - Economic magnitude of the results

6. Evaluation (relevant mostly for forecasting projects)
   - If possible, quantify the success of your out-of-sample forecasts or portfolio performance
   - interpret the findings from the firm's point of view`

7. Conclusion
   - What did you learn about the problem you set out to analyze?
   - What are your key findings (positive as well as negative)?
   - How useful are your findings to decision makers?
   - What were the most surprising/headline results from the project?

8. References used in the project

There is no prescribed style. Teams may follow the format of popular practitioner journals like *Financial Analysts Journal, Journal of Portfolio Management*, or *Risk* or a scholarly finance journal like the *Journal of Finance* or *Journal of Financial Economics*.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/manual/appendices/appendix2.pdf

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.